

Literature Status: Unconditional Bases and SCD Spaces

Run `fa_banach_001`, agent lane 8

2026-06-14

Status

literature_already_answered. The open question extracted from arXiv:2311.03064 is answered negatively in the later paper arXiv:2603.12947. This packet records provenance only; it is not an original proof from the run.

Source Question

The source paper is Johann Langemets, Marcus Loo, Miguel Martin, and Abraham Rueda Zoca, *Slicely countably determined points in Banach spaces*, arXiv:2311.03064. In the introduction, immediately after recalling that the unit ball of every Banach space with a one-unconditional basis is SCD, the authors state:

“It is an open question whether every Banach space with an unconditional basis is an SCD space.”

Supporting Answer

The supporting paper is Marcus Loo and Yoel Perreau, *A Banach space with an unconditional basis which is not slicely countably determined*, arXiv:2603.12947. Its Introduction explicitly restates the same problem as Question 1.2:

“Let X be a Banach space with an unconditional basis. Is X necessarily an SCD space?”

The paper then says that it provides a negative answer to Question 1.2. Its Theorem 1.3 states that the positive unit ball $B_{X_T}^+$ of the binary tree space is not an SCD set and, in particular, X_T is a Banach space with an unconditional basis which fails to be SCD.

Scope

The answer is negative for the exact unconditional-basis SCD-space question. The supporting authors explicitly know they are answering this question: they state it as Question 1.2 and identify Theorem 1.3 as the negative answer. No claim is made here about the later paper’s separate open questions, such as countable π -bases or the existence of a Banach space with an unconditional basis whose unit ball has no SCD point.

Search Evidence

The source arXiv id, title keywords, and SCD/unconditional-basis keywords were searched in the run's cheap indexes. No existing packet recorded the arXiv:2311.03064 source-question relation. A web search for "every Banach space with an unconditional basis" "SCD" found arXiv:2603.12947, whose abstract and introduction state the negative answer. The run already contains a separate packet about a different question from arXiv:2603.12947, namely k -unconditional renormings; this packet records only the later-paper answer to the arXiv:2311.03064 unconditional-basis question.

References

- [1] J. Langemets, M. Loo, M. Martin, and A. Rueda Zoca, *Slicely countably determined points in Banach spaces*, arXiv:2311.03064, 2023. <https://arxiv.org/abs/2311.03064>.
- [2] M. Loo and Y. Perreau, *A Banach space with an unconditional basis which is not slicely countably determined*, arXiv:2603.12947, 2026. <https://arxiv.org/abs/2603.12947>.